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Lesson 5: Experimental Methodology

Technologies for Artificial Intelligence

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Before we start: lesson 4, concluded

- First, lesson 4’s last slides: receiver operating characteristic (ROC) curves, the area under the curve (AUC), the threshold from costs, notebook 3
- Exercise 4 is set at their end; exercises 4 and 5 are both discussed on 13 November
- Then lesson 5, with a few parts left for self-study, marked (optional)

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We broke a promise on purpose

- Lesson 4 chose a threshold **by looking at the test set**
- Lesson 1 forbade exactly that
- Today we pay the debt

Today: how to tell whether it works

- One train/test split gives one number; another split gives another. How far apart?
- Cross-validation: what it estimates, and what it does not
- Bias, variance, and the part of the error no model can remove
- Learning curves: will more data help, or is a different model needed?
- Leakage that survives cross-validation

A test score is a measurement

- You would not report a poll of forty people to three decimals
- Holding data out makes the estimate **unbiased**: right on average, over all the splits you might have drawn
- It does nothing to make it **precise**: close to the truth on the one split you did draw

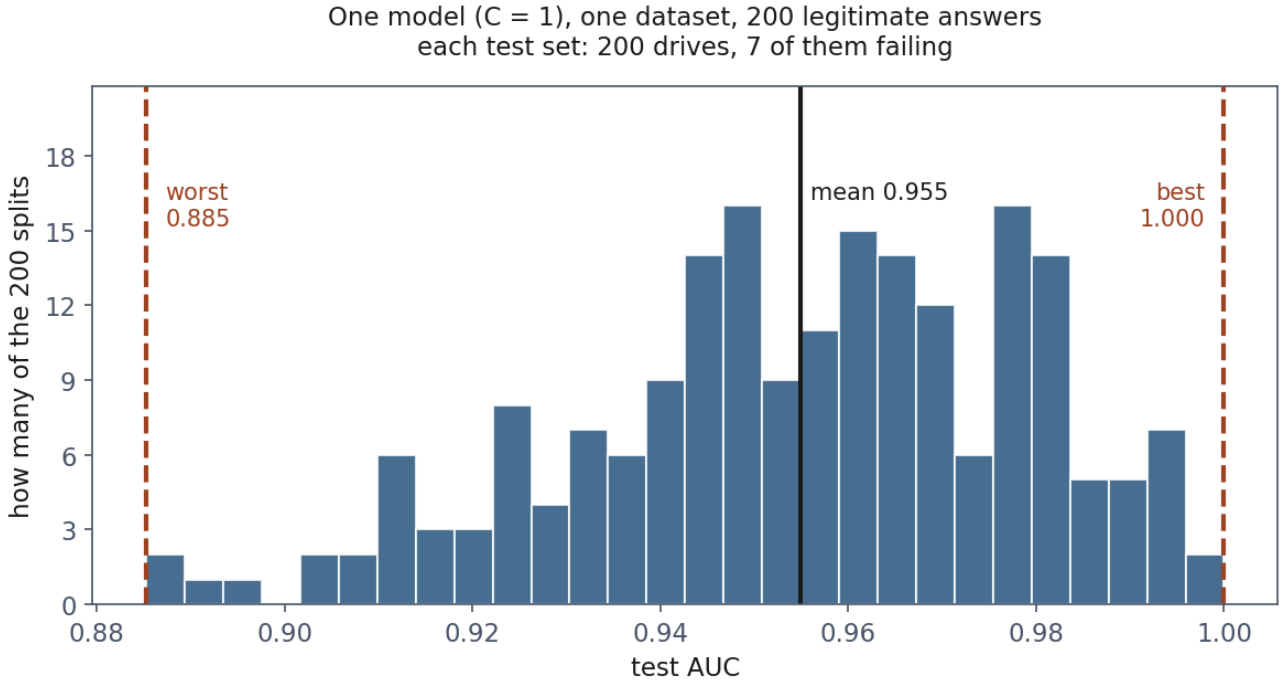
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The fleet, cut down

- Lesson 4’s drives, reduced to **800**, of which **29 fail** (3.6%)
- A few hundred rows with a few dozen positives is the ordinary case
- In code: lesson 4’s model, `make_pipeline(StandardScaler(), LogisticRegression())`, on `train_test_split(X, y, test_size=0.25, stratify=y)`, scored by `roc_auc_score`

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200 honest splits, one dataset



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Why so wide?

- Each test set takes 25% of the 29 failures: $0.25 \times 29 = 7.25$, so **7 failures** among 200 drives
- The AUC asks only how those 7 failing drives are ranked against the 193 healthy ones: $7 \times 193 = 1{,}351$ pairs
- One failing drive slipping below 20 more healthy ones costs $20 \div 1{,}351 = 0.015$ of AUC

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Parameters and hyperparameters

- **Parameters**: estimated by the fit, w and b
- **Hyperparameters**: not estimated by the fit: penalty λ (lesson 3), threshold (lesson 4), neighbours (lesson 6), tree depth (lesson 7)
- Minimising training error would always pick $\lambda = 0$, so they are chosen by **comparing** models on data the fit did not see
- scikit-learn's C plays the role of $1/\lambda$: the smaller C, the stronger the penalty

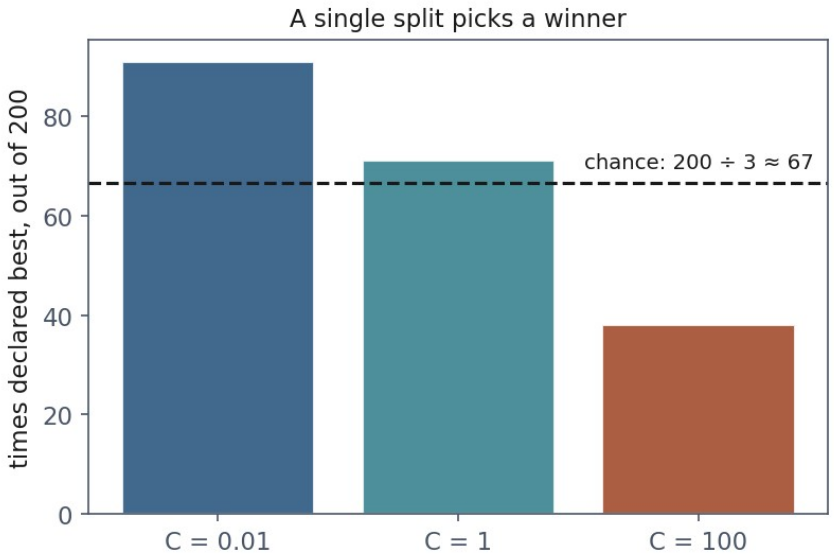
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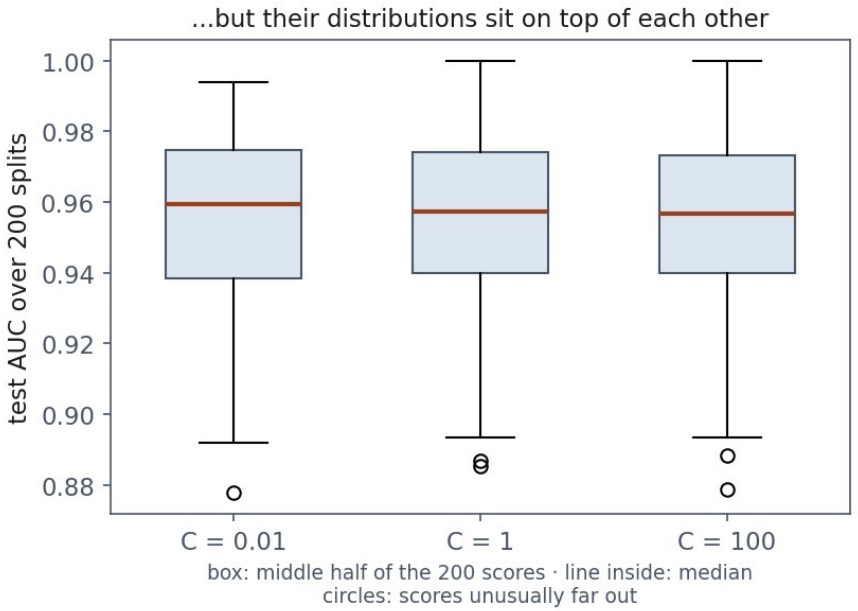
On each of slide 7’s 200 splits, three models compete

Model (C = 1, the default, is slide 7’s)	Mean test AUC over the 200 splits	Splits it won (chance: 200 ÷ 3 ≈ 67)
C = 0.01	0.9549	91
C = 1	0.9549	71
C = 100	0.9544	38

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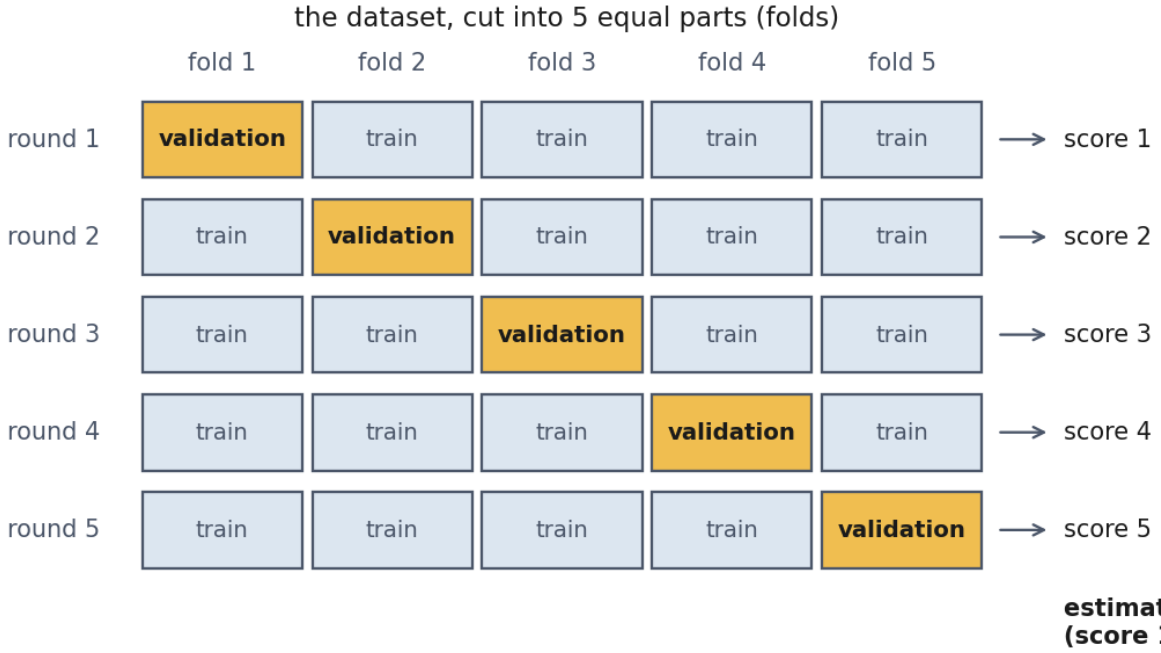
The same three models: who wins, and how they score





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k-fold cross-validation



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What it actually estimates

- The performance of a **procedure** (the whole recipe), not of one fitted model
- Slightly **pessimistic**: with 5 folds, each model trains on 4/5 = 80% of the data
- Seed 0: the mean of the five fold AUCs is **0.9514**
- In code: `cross_val_score(model, X, y, cv=StratifiedKFold(5, shuffle=True), scoring="roc_auc")`, one score per fold

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Report the spread, but not as a confidence interval

- With 5 folds each training set is 4 folds, and any two share 3 of them: **3/4** of their rows
- So the fold scores are correlated, and the usual standard error s ÷ √k (s the standard deviation of the k fold scores) comes out **too small**
- Report the mean and s, call s a spread, and treat a small difference between two models as unresolved

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Stratify when a class is rare

Fold (seed 3)	KFold: folds cut at random	StratifiedKFold: each fold keeps 3.6% failures
1	7 failures of 160	5 of 160
2	**1 failure of 160**	6 of 160
3	7 of 160	6 of 160
4	5 of 160	6 of 160
5	9 of 160	6 of 160

How much does it help? (optional)

Test AUC, C = 1, seeds 0-39	Single 75/25 split	5-fold cross-validation
worst	0.9119	0.9439
best	1.0000	0.9588
mean	0.9600	0.9532
best − worst	0.0881	0.0149
standard deviation	0.0221	0.0032

How many folds? (optional)

- **k = 5 or 10** in practice
- Larger k: less pessimistic, more fits to run, fold scores more correlated
- k = m (m the number of rows) is **leave-one-out**: every row is a validation fold on its own

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When the rows are not independent

- Suppose the table held ten readings per drive (lesson 4’s has one row per drive)
- A random split puts nine of a drive’s readings in training and one in test
- The model recognises the drive, not the failure

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Split by the thing you must generalise to

- If one drive (or patient, or customer) has several rows: **GroupKFold** keeps all its rows in the same fold
- If the model must predict the future: **TimeSeriesSplit** trains on the past and validates on what comes after it

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Notebook 1, live

section, and its slides	you call (→ the number to come back with)
two honest experiments, 5-6	`train_test_split(..., random_state=seed)`, `make_pipeline(StandardScaler(), LogisticRegression())`, `roc_auc_score` → seed 3 gives 1.000
how wide, 7-8	the same, 200 times → 0.885 to 1.000
choosing on one split, 9-11	three values of C, a winner per split → 91, 71 and 38 wins
k-fold by hand, 12-14	`StratifiedKFold`, then `cross_val_score` → mean 0.9514, standard deviation 0.0187, by hand and by scikit-learn alike
stratify, 15-16	`KFold` against `StratifiedKFold` → a fold with 1 failure; best − worst 0.0881 against 0.0149

Break

- Ten minutes

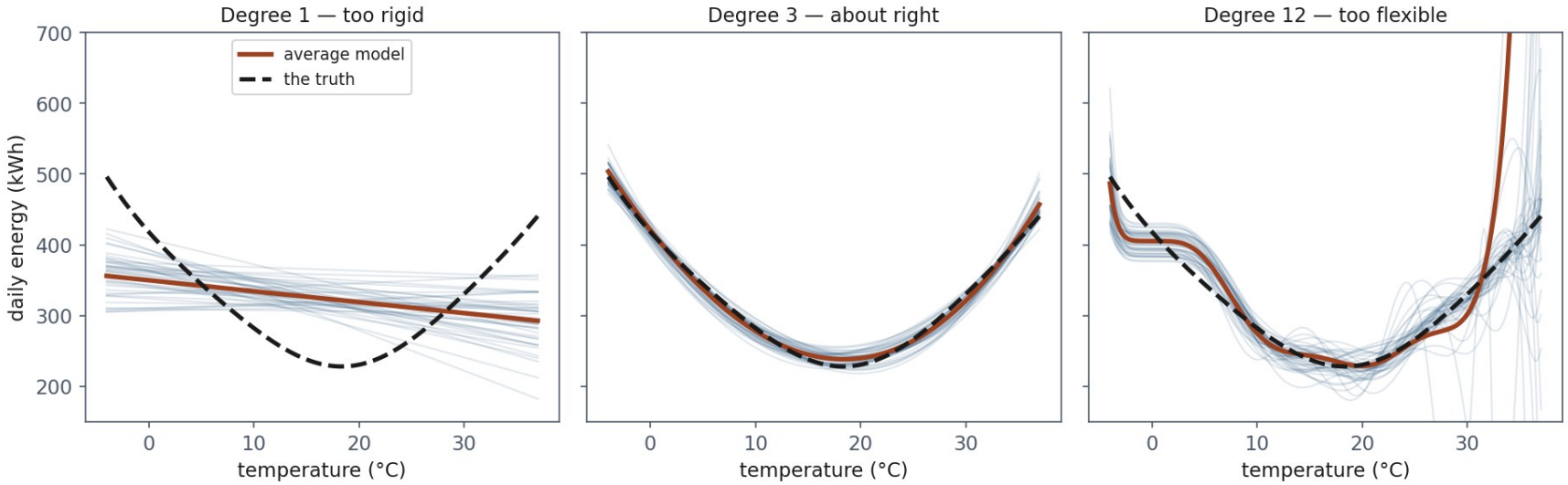
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Where does the error come from?

- **Bias²**: (average prediction − truth)²: the average model is wrong
- **Variance**: the spread of the models around their average
- **Noise** σ^2: the target itself is uncertain
- Lesson 3's energy curve, 300 training sets of 25 days, each fitted with `make_pipeline(PolynomialFeatures(degree), LinearRegression())`

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Three hundred parallel universes



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The decomposition

$$\mathrm{E}\left[(y - \hat{f})^2\right] = \mathrm{bias}^2 + \mathrm{variance} + \sigma^2$$

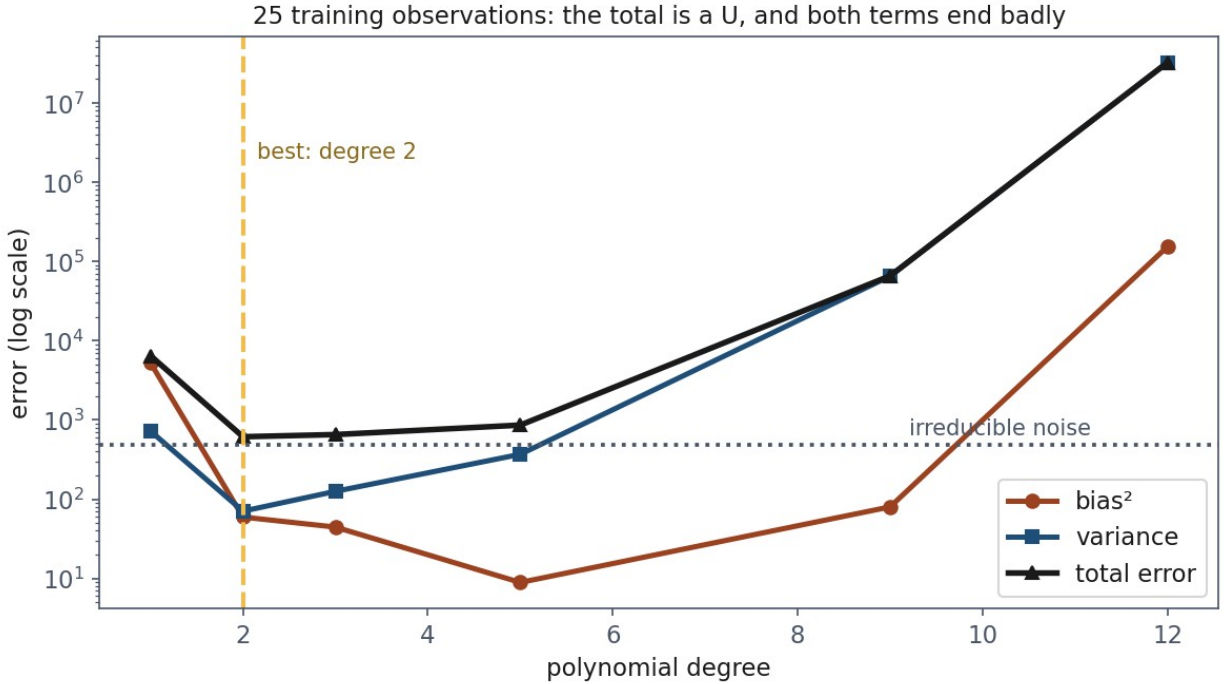
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Measured, on 25 observations

Degree	bias²	Variance	Noise	Total
1	5,250.8	723.8	484.0	6,458.6
2	59.6	70.8	484.0	**614.3**
5	8.9	368.9	484.0	861.8
12	153,537.3	32,174,344.9	484.0	32,328,366.2

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The picture everyone has seen, now measured



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The floor you cannot go below (optional)

- Noise variance is **484** = 22^2, the noise put into the data, at every complexity
- A reported error well below a known floor means **contamination**, not excellence

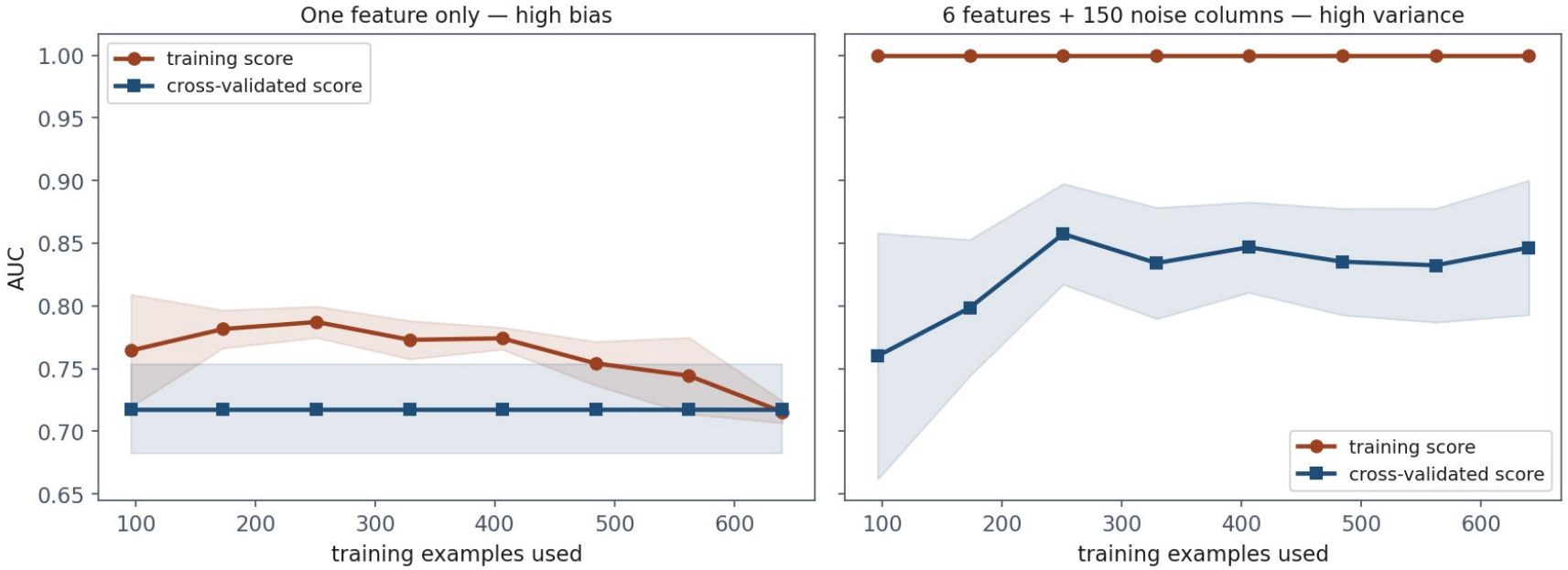
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The best model depends on how much data you have (optional)

Degree	m = 25	m = 60	m = 150
2	**614.3**	570.6	555.3
5	861.8	**540.9**	**506.9**
12	32,328,366.2	7,626.9	899.0

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Learning curves: the same diagnosis, without knowing the truth



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Which fix to buy

- A learning curve: training and cross-validated score against training-set size,
  `learning_curve(model, X, y, cv=folds, scoring="roc_auc")`
- Curves **meet low** → bias. More data will **not** help
- Wide gap, validation **still rising** → more data **will**
- Wide gap, validation **flat** → regularise, drop features. **Ours is this one**

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Notebook 2, self-study (optional)

section, and its slides	you call (→ the number to come back with)
300 parallel universes, 22-23	`PolynomialFeatures` then `LinearRegression`, in `make_pipeline`, fitted 300 times
the three parts measured, 24-27	NumPy on the 300 predictions → sum and measured error agree to 2.73e-12
more data moves the answer, 28	the same at 25, 60 and 150 days → best degree 2, then 5
learning curves, 29-30	`learning_curve` with `StratifiedKFold` → 0.715 against 0.718; 1.000 against 0.847

Now: what cross-validation does not protect you from

- 2,000 columns of **pure random noise**, 800 rows, 29 positives
- No relationship to the label whatsoever
- The honest score of any model on it is **0.500**
- The plan: keep the 10 columns whose F-test against the label scores highest, `SelectKBest(f_classif, k=10)`, then cross-validate

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The code almost everyone writes

```
sel = SelectKBest(f_classif, k=10)
sel.fit(noise, y)
X = noise.loc[:, sel.get_support()]
cross_val_score(model, X, y, cv=cv)
```

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0.931

- Cross-validated AUC **0.931** on data with no signal
- Four of five folds between 0.92 and 0.98

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Fix it, and look what happens

- Selection moved **inside** the pipeline, refitted per fold: `Pipeline([("select", SelectKBest(f_classif, k=10)), ("scale", ...), ("model", ...)])`
- 20 different fold seeds: **0.658** on average, 0.576 to 0.759
- The truth is still 0.500

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Why it does not return to 0.5

- With 2,000 columns and 800 rows, some columns match the label **by accident**
- That accident belongs to the **sample**, not to the split
- Every fold contains it

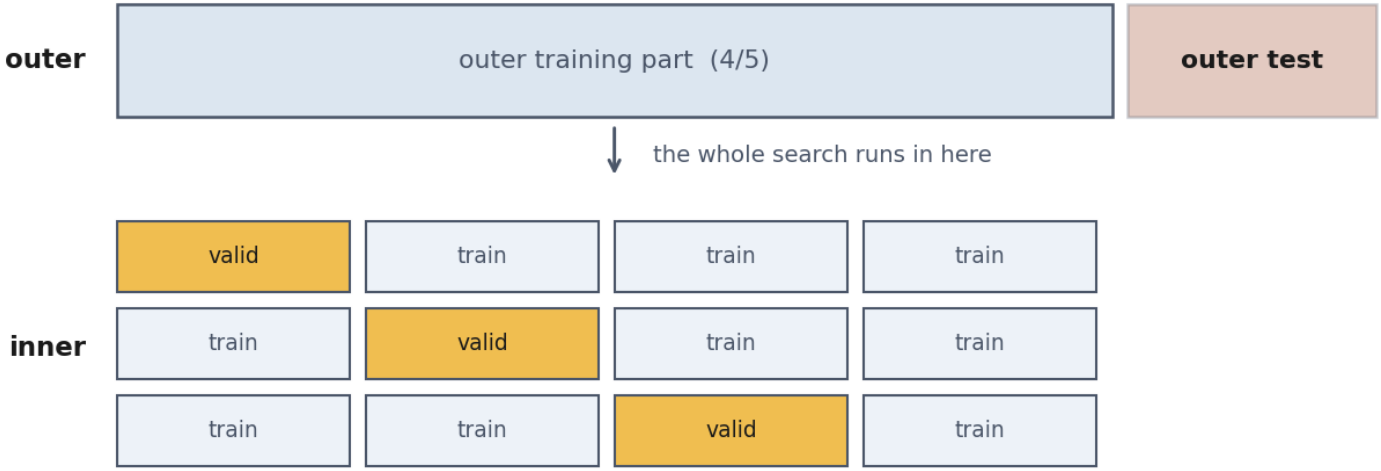
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Hyperparameter search is the same problem

`GridSearchCV`, 5 values of k × 5 of C	AUC
best of 25 candidates, `best_score_`	**0.7999**
average candidate	0.7265
nested cross-validation, next slide	0.6699
the truth, by construction	0.5

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Nested cross-validation

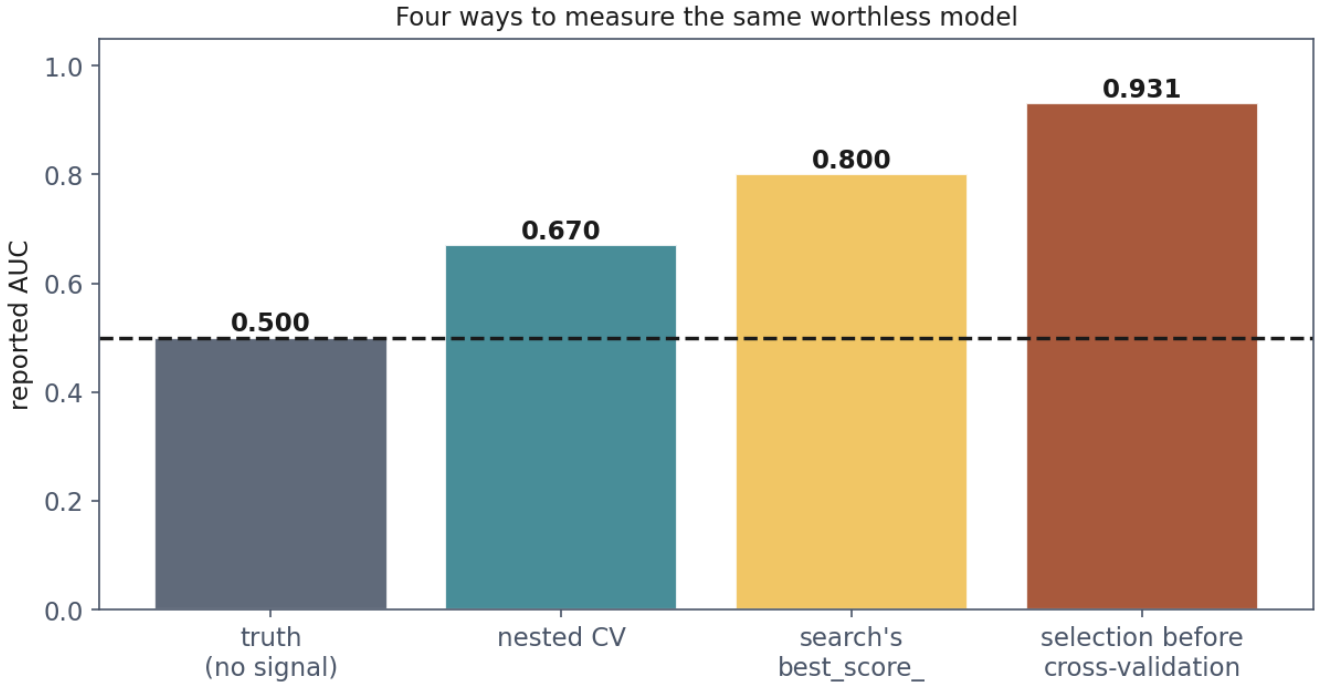


The inner loop may overfit itself as much as it likes.

The outer test block was never part of it: that is the number you report.

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Four ways to measure the same worthless model



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The debt, paid

Slide 6’s 800 drives: fit on 450, choose on 150, report on 200 (7 failures)	Missed × 2,600 + false alarms × 140	Cost on the test set
0.50, the default	4 × 2,600 + 1 × 140	10,540 EUR
0.050, chosen on **validation**	1 × 2,600 + 15 × 140	**4,700 EUR**
0.171, chosen on **test**	1 × 2,600 + 5 × 140	3,300 EUR

Three sets, three jobs

- **Train**: fit the parameters
- **Validation**: make every choice: model, hyperparameters, threshold
- **Test**: report. Touched **once**

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A fixed seed buys repeatability, not stability (optional)

- 30 seeds, all perfectly reproducible: **0.913 to 1.000**
- `AUC 1.000 (random_state=3)` is reproducible **and** misleading

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What to fix, and what to write down (optional)

- Every `random_state`, and `np.random.default_rng(seed)`
- **Library versions**: defaults change between releases
- What you could **not** fix: threads, graphics processing unit (GPU) non-determinism

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Notebook 3, live

section, and its slides	you call (→ the number to come back with)
the catastrophe, 32-34	`SelectKBest(f_classif, k=10)` on every row, then `cross_val_score` → 0.931
selection inside the fold, 35-36	`Pipeline` with the selector as its first step → 0.658
the search, 37	`GridSearchCV`, `best_score_` → 0.7999
nested, 38-39	`cross_val_score(GridSearchCV(...), cv=outer)` → 0.6699
the threshold, honestly, 40-41	`train_test_split` twice → 4,700 EUR against 10,540
seeds, 42-43	30 values of `random_state` → 0.913 to 1.000

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What to put in a report

- The metric, the **spread**, and how it was estimated
- How many **positives** the test set held
- Every seed, and every library version
- Which choices were made on which data

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What to take away

- **A test score is a measurement.** 0.885 to a perfect **1.000**, same model
- **Choosing on one split chooses noise**
- **bias² + variance + noise** is an identity, and the noise floor never moves
- **Curves that meet low mean bias**: more data will not help
- **Cross-validation protects against fold leakage, nothing more**

Homework

- **Exercise 5**, discussed at the start of **Friday 13 November**
- A result that is too good. Find out why, then produce the honest number